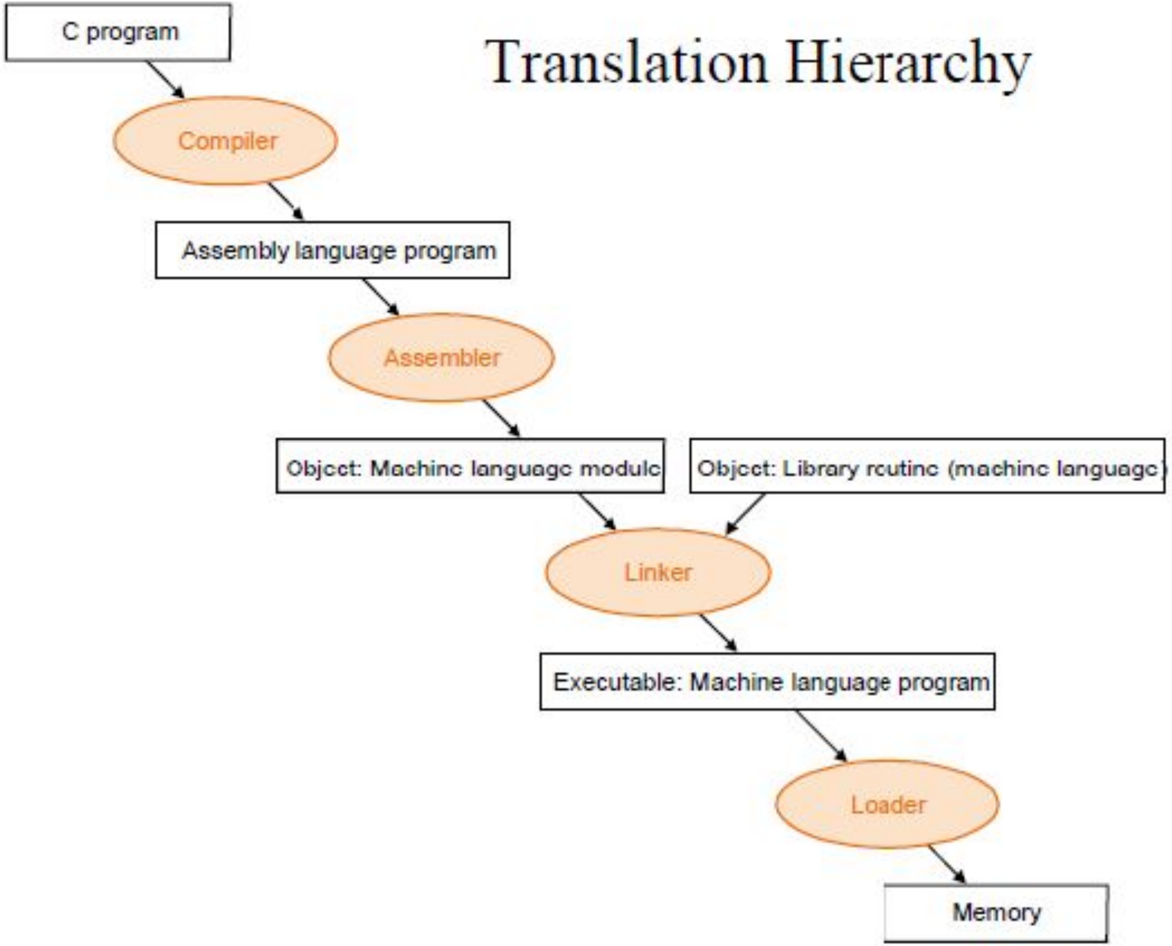
A Beginners guide to Assembly

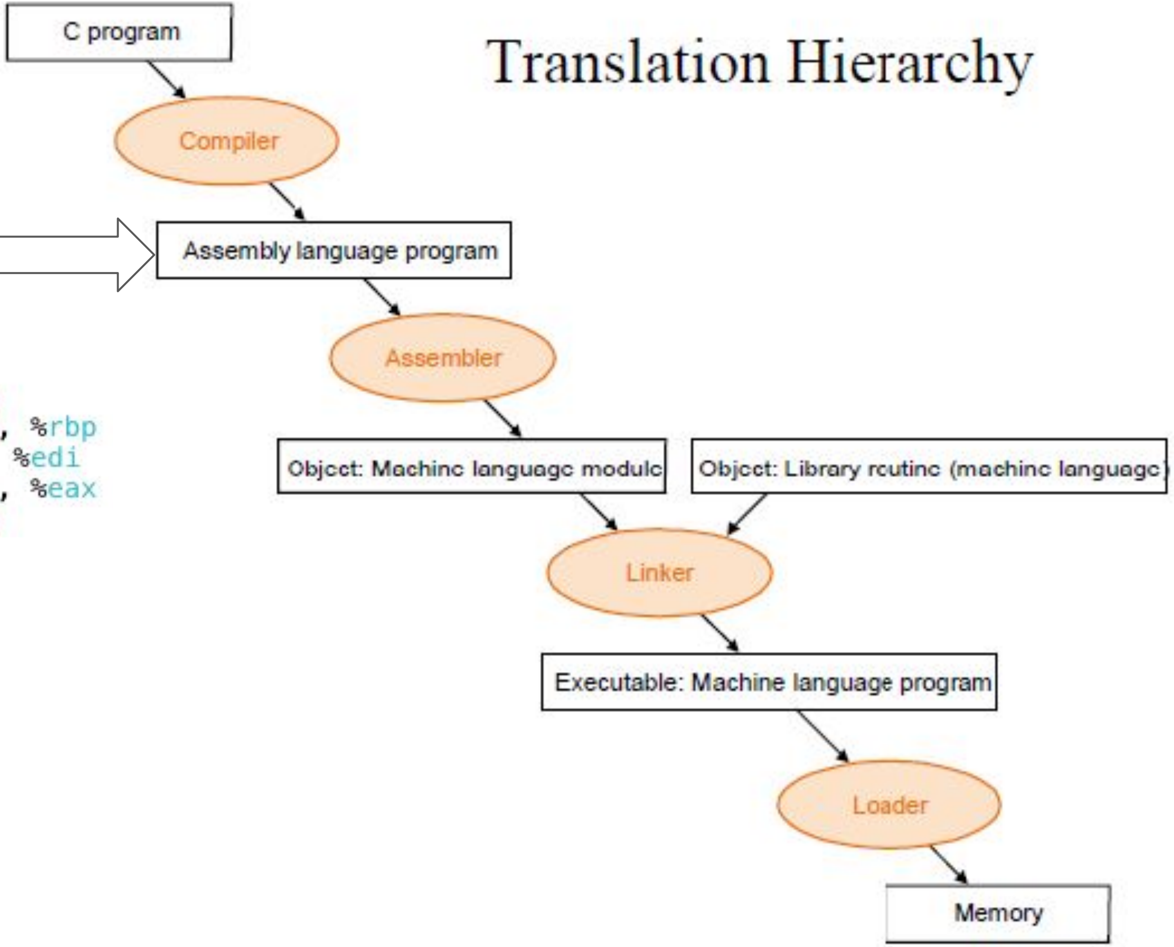
By Tom Glint and Rishiraj
CS301 | Fall 2020

Contributors: Varun and Shreyas

Translation Hierarchy

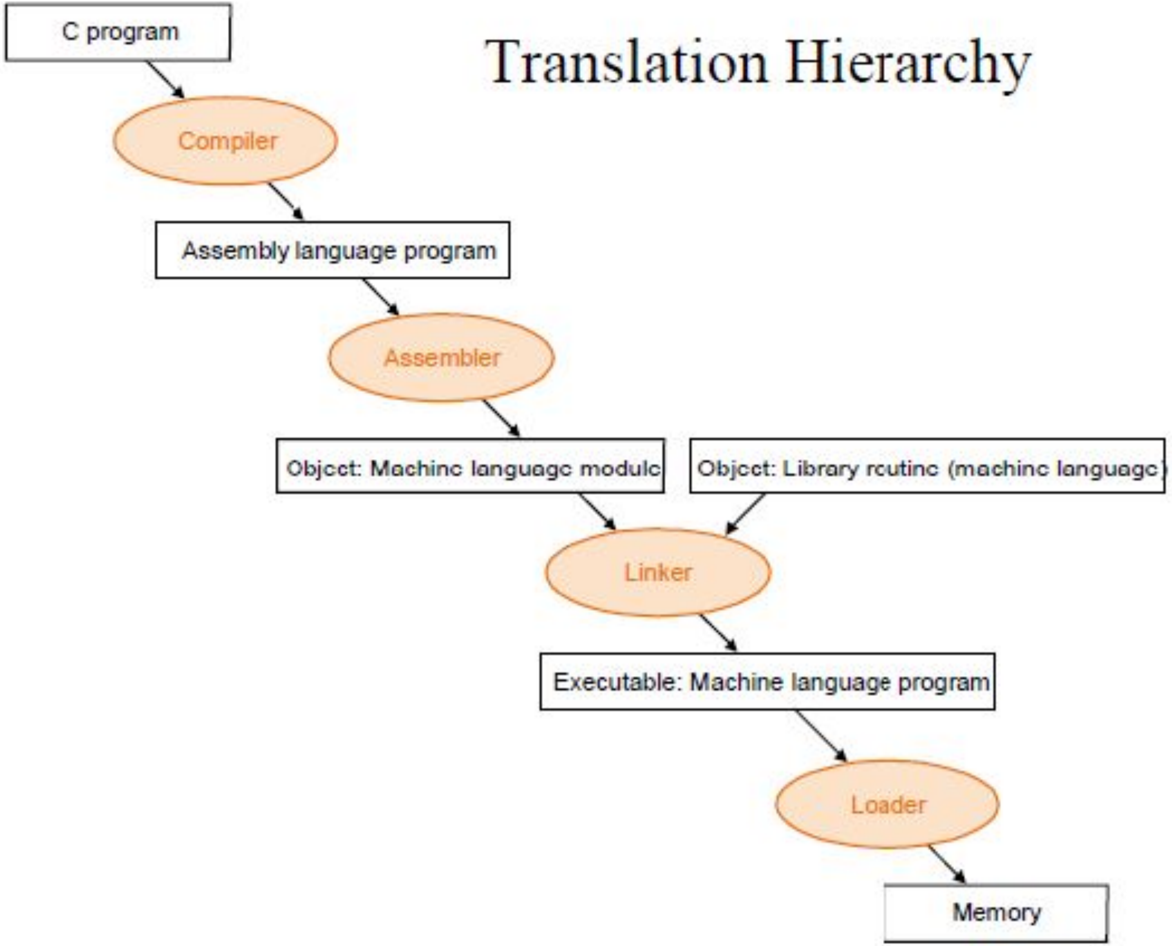


Translation Hierarchy




```
pushq %rbp
movq  %rsp, %rbp
addl  $42, %edi
movl  %edi, %eax
popq  %rbp
```

Translation Hierarchy



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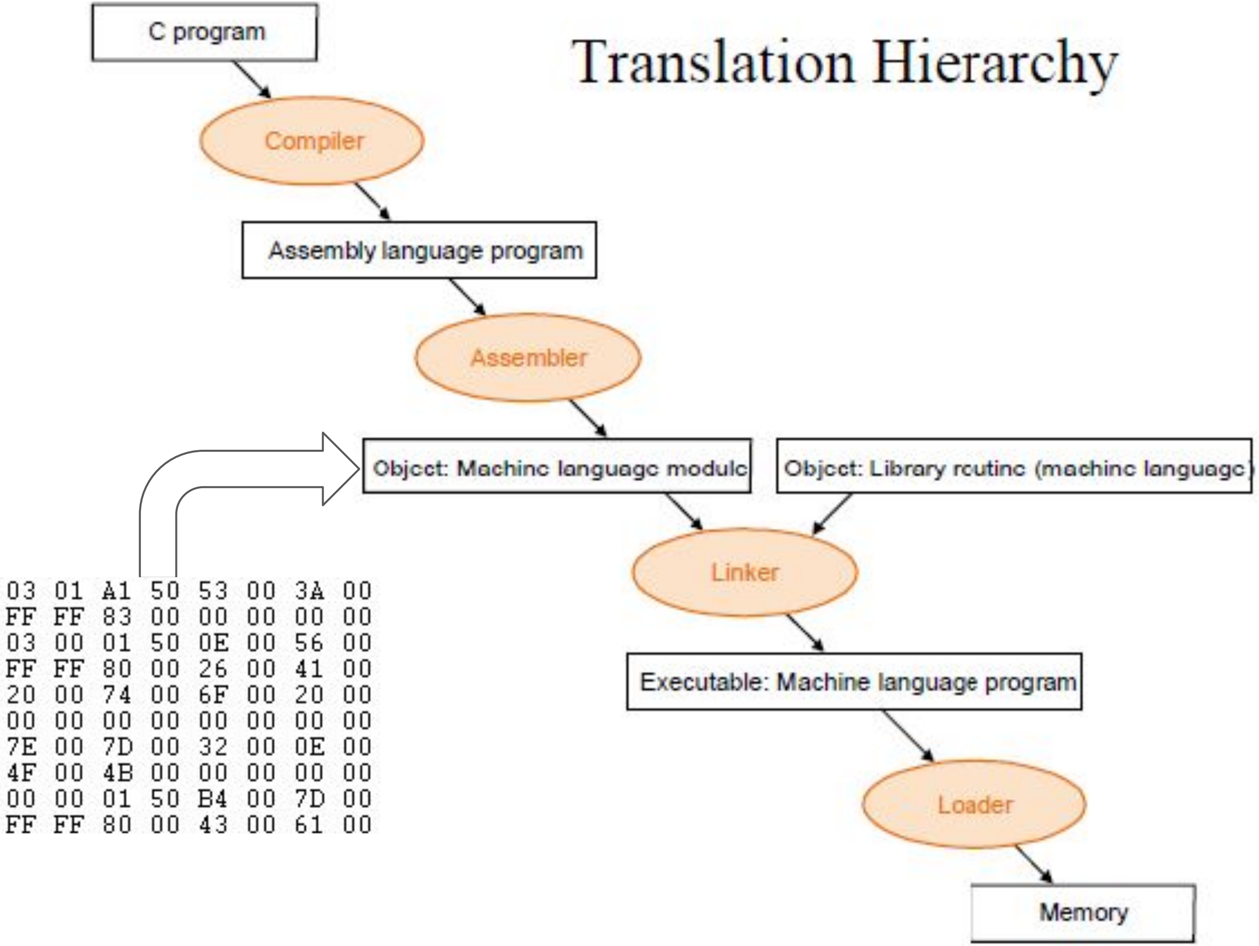

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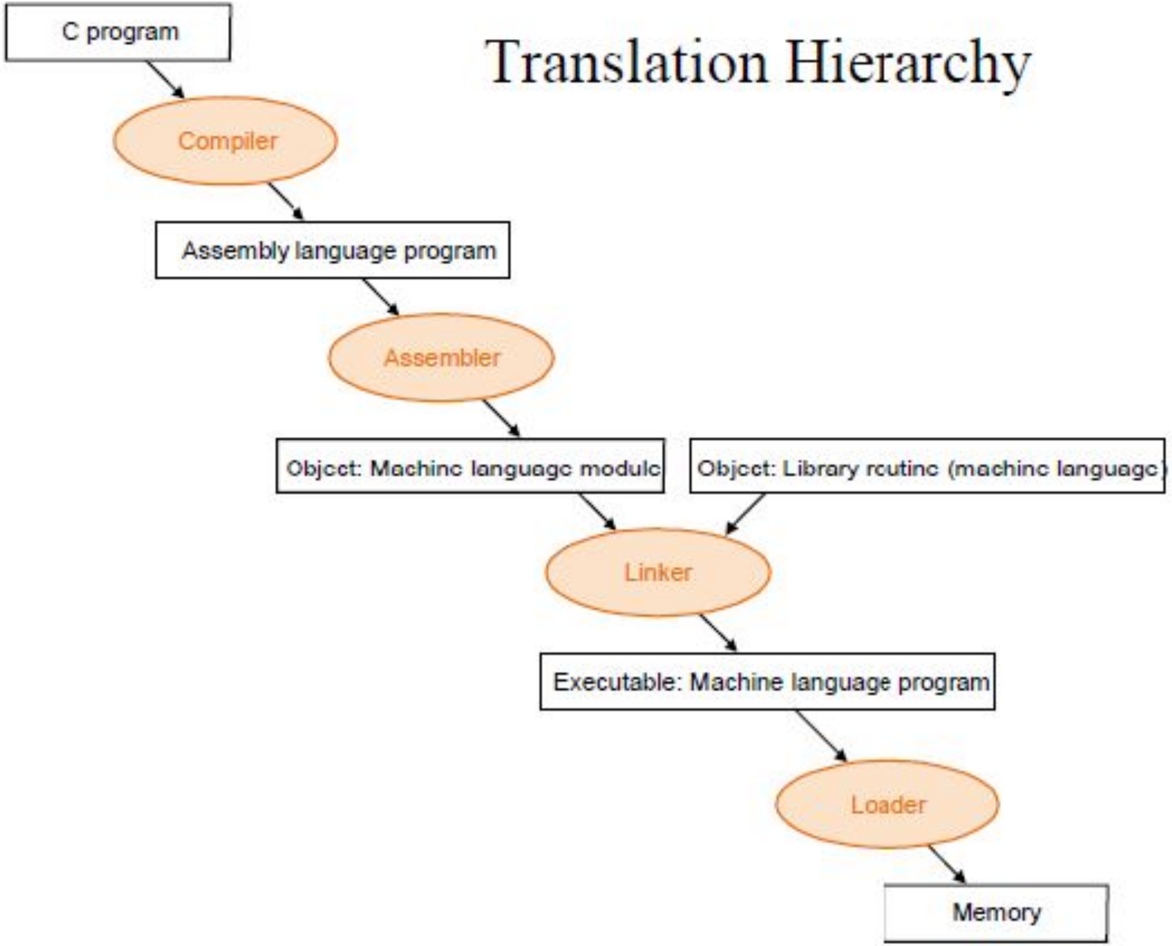

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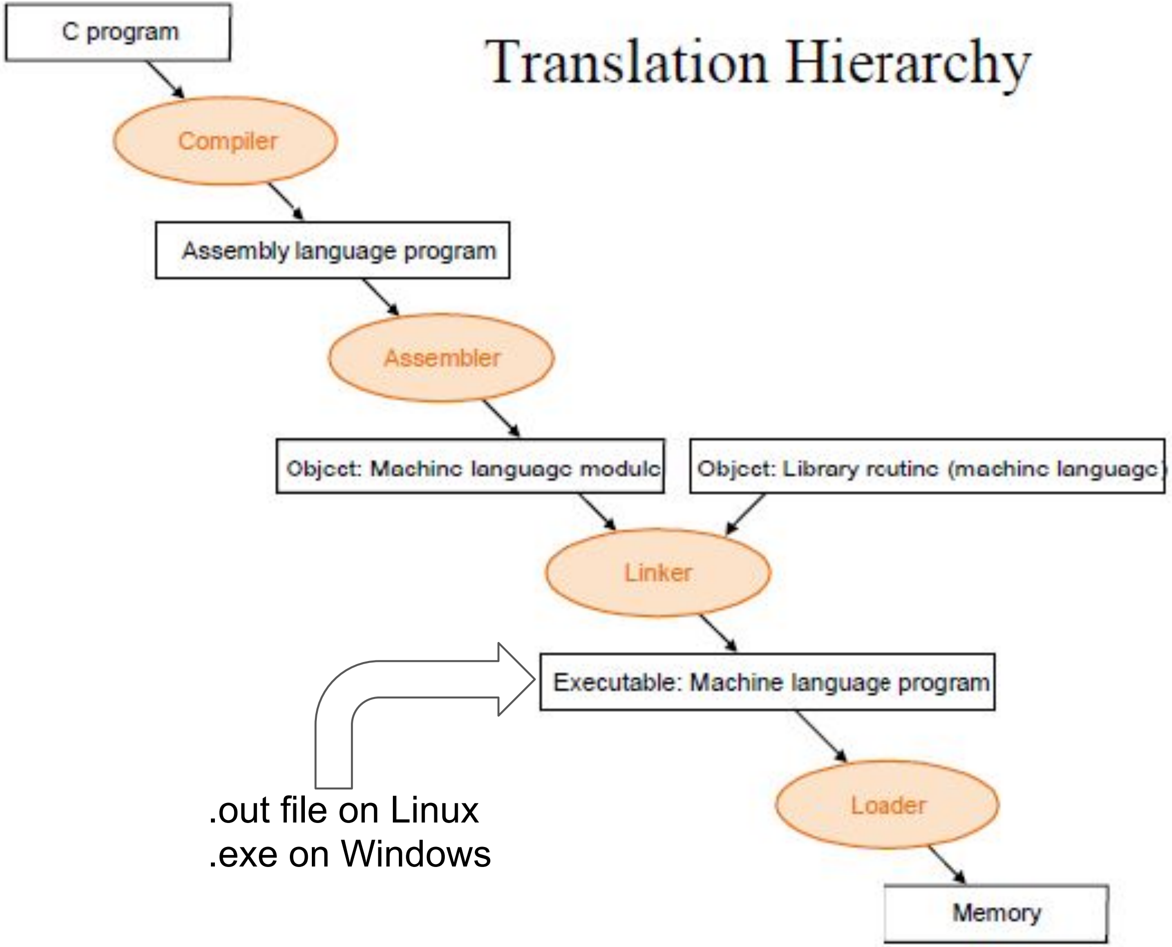

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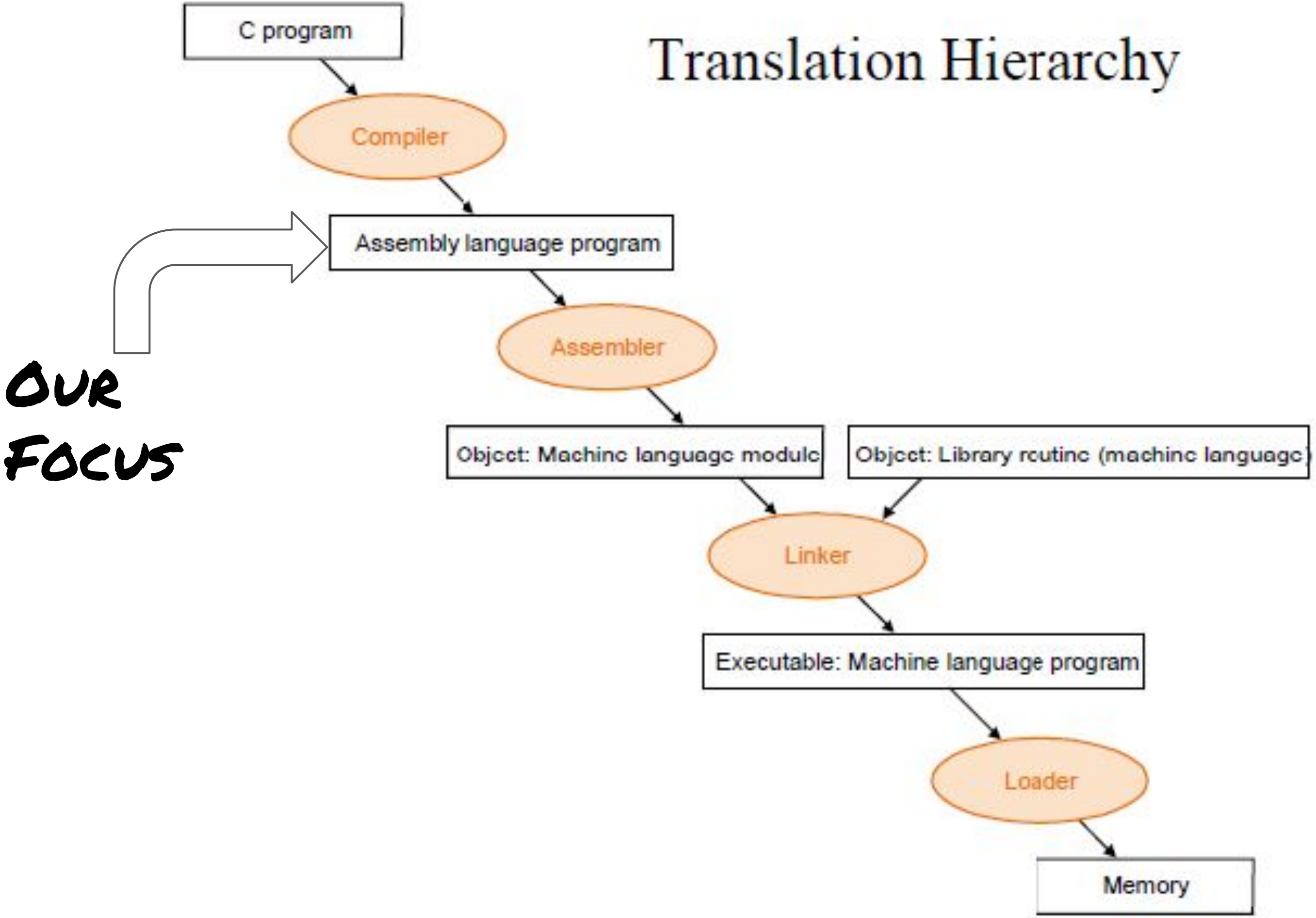

Prominent ISAs

intel x86

RISC-V

ARM

IIT-Madras Develops 'India's First Microprocessor', Shakti

By Indo-Asian News Service | Updated: 2 November 2018 16:03 IST

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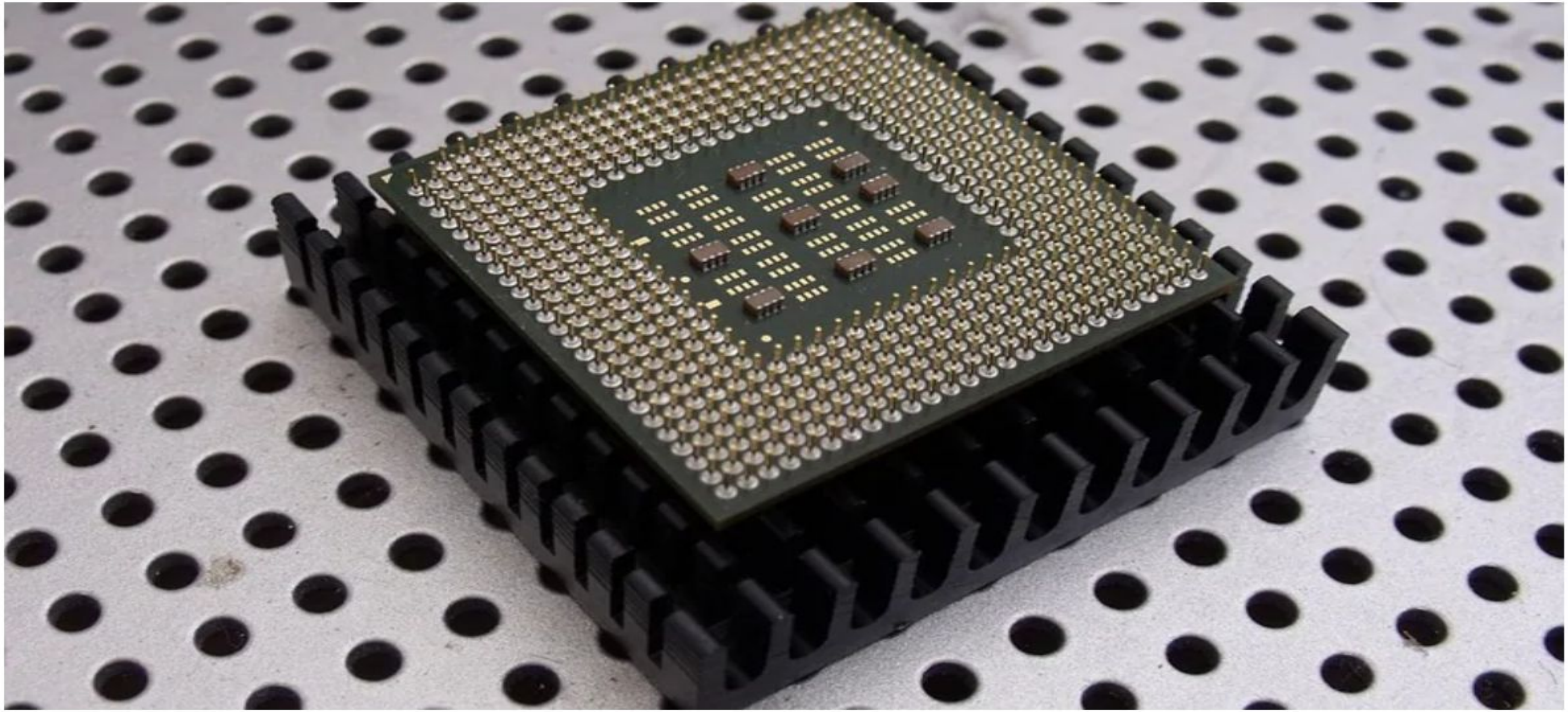

An intriguing Example!

```c
#include<stdio.h>

int main(){
	int x = 3000,y;
	x = x + 3;
	y = 100;
	return y;
}
```

```asm
main:
	pushq	%rbp
	movq	%rsp, %rbp
	movl	$3000, -8(%rbp)
	addl	$3, -8(%rbp)
	movl	$100, -4(%rbp)
	movl	-4(%rbp), %eax
	popq	%rbp
	ret
```

Some Basics

- % - indicates register names. Example : %rbp
- $ - indicates constants Example : $100
- Accessing register values:
  - %rbp   : Access value stored in register rbp
  - (%rbp)   : Treat value stored in register rbp as a pointer. Access the value stored at address pointed by the pointer. Basically *rbp
  - 4(%rbp) : Access value stored at address which is 4 bytes after the address stored in rbp. Basically *(rbp + 4)

An intriguing Example!

```c
#include<stdio.h>

int main(){
        int x = 3000,y;
        x = x + 3;
        y = 100;
        return y;
}
```

```
main:
        pushq   %rbp
        movq    %rsp, %rbp
        movl    $3000, -8(%rbp)
        addl    $3, -8(%rbp)
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        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

An intriguing Example!

```c
#include<stdio.h>

int main(){
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}
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        pushq   %rbp
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        movl    $3000, -8(%rbp)
        addl    $3, -8(%rbp)
        movl    $100, -4(%rbp)
        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

For each function call, new space is created on the stack to store local variables and other data. This is known as a stack frame. To accomplish this, you will need to write some code at the beginning and end of each function to create and destroy the stack frame

An intriguing Example!

```c
#include<stdio.h>

int main(){
        int x = 3000,y;
        x = x + 3;
        y = 100;
        return y;
}
```

```
main:
        pushq   %rbp
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        movl    $3000, -8(%rbp)
        addl    $3, -8(%rbp)
        movl    $100, -4(%rbp)
        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

rbp is the frame pointer. In our code, it gets a snapshot of the stack pointer (**rsp**) so that when rsp is changed, local variables and function parameters are still accessible from a constant offset from **rbp**.

An intriguing Example!

```c
#include<stdio.h>

int main(){
        int x = 3000,y;
        x = x + 3;
        y = 100;
        return y;
}
```

```
main:
        pushq   %rbp
        movq    %rsp, %rbp
        movl    $3000, -8(%rbp)
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        movl    $100, -4(%rbp)
        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

move immediate value 3000 to (%rbp-8)

An intriguing Example!

```c
#include<stdio.h>

int main(){
        int x = 3000,y;
        x = x + 3;
        y = 100;
        return y;
}
```

→

```
main:
        pushq   %rbp
        movq    %rsp, %rbp
        movl    $3000, -8(%rbp)
        addl    $3, -8(%rbp)
        movl    $100, -4(%rbp)
        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

add immediate value 3 to (%rbp-8)

An intriguing Example!

```c
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int main(){
        int x = 3000,y;
        x = x + 3;
        y = 100;
        return y;
}
```

```
main:
        pushq   %rbp
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        movl    $3000, -8(%rbp)
        addl    $3, -8(%rbp)
        movl    $100, -4(%rbp)
        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

Move immediate value 100 to (%rbp-4)

An intriguing Example!

```c
#include<stdio.h>

int main(){
        int x = 3000,y;
        x = x + 3;
        y = 100;
        return y;
}
```

```
main:
        pushq   %rbp
        movq    %rsp, %rbp
        movl    $3000, -8(%rbp)
        addl    $3, -8(%rbp)
        movl    $100, -4(%rbp)
        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

Move (%rbp-4) to auxiliary register

An intriguing Example!

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#include<stdio.h>

int main(){
        int x = 3000,y;
        x = x + 3;
        y = 100;
        return y;
}
```

```
main:
        pushq   %rbp
        movq    %rsp, %rbp
        movl    $3000, -8(%rbp)
        addl    $3, -8(%rbp)
        movl    $100, -4(%rbp)
        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

Pop the base pointer to restore state

An intriguing Example!

```c
#include<stdio.h>

int main(){
        int x = 3000,y;
        x = x + 3;
        y = 100;
        return y;
}
```

```
main:
        pushq   %rbp
        movq    %rsp, %rbp
        movl    $3000, -8(%rbp)
        addl    $3, -8(%rbp)
        movl    $100, -4(%rbp)
        movl    -4(%rbp), %eax
        popq    %rbp
        ret
```

The calling convention dictates that a function's return value is stored in %eax, so the above instruction sets us up to return y at the end of our function.

Operation Suffixes

- b = byte (8 bit)
- s = single (32-bit floating point)
- w = word (16 bit)
- l = long (32 bit integer or 64-bit floating point)
- q = quad (64 bit)
- t = ten bytes (80-bit floating point)

How to get assembly code?

Two ways:

- `While Compiling`
  - `Use` **`-S flag with gcc`**`. WIll create a .s file containing assembly`

- `Using Binary`
  - `Use` **`objdump`**`. Will show the assembly in terminal.`

Understanding the output

- The output will have assembly, but there is more information!
- You will see lots of Directives like:
  - .file
  - .text
  - .global name

Understanding the output

- The output will have assembly, but there is more information also!.
- You will see lots of Directives like:
    - .file
    - .text
    - .global name

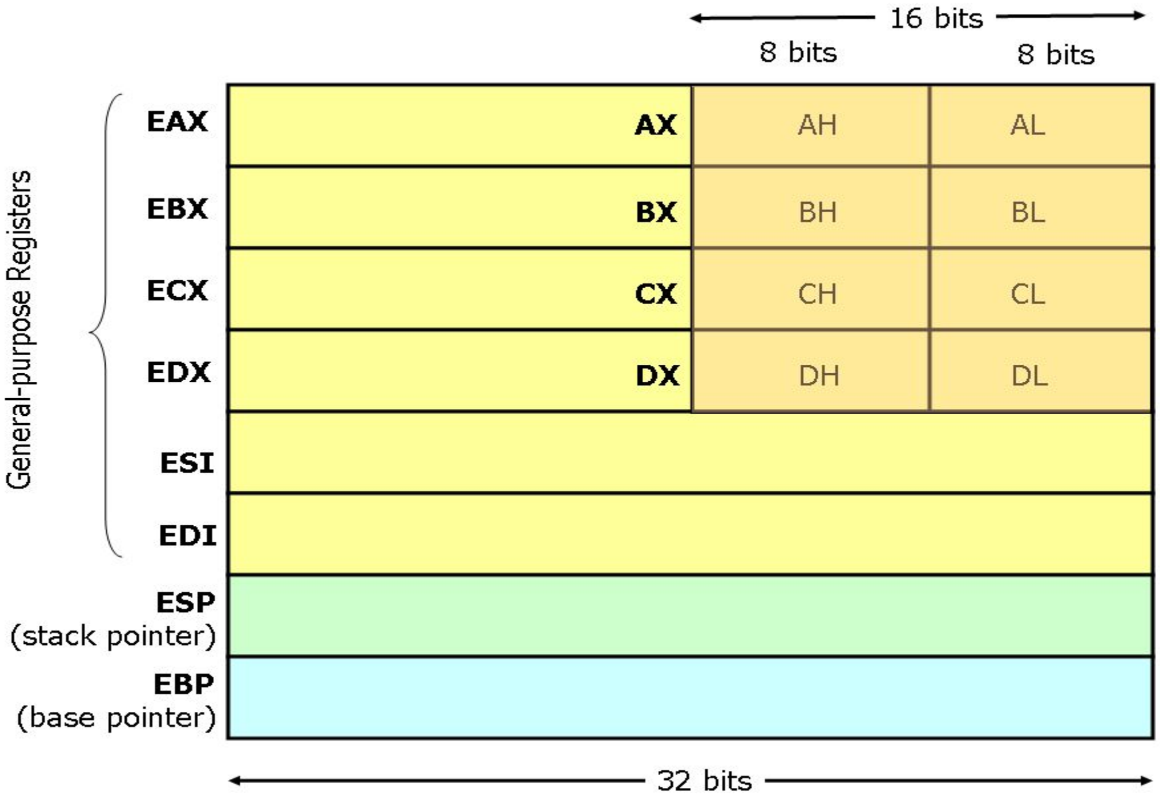
137

To disable these, use the gcc option

```
-fno-asynchronous-unwind-tables
```

Note, I know this is a really old thread, but this is the top result on google for cfi_startproc, so many people probably come here to disable that output.

x86 Register Set



x86 Register Set : A few more

- Registers starting with “r”
  - Same as “e” registers but 64 bits wide
- EIP : The Instruction Pointer or the Program Counter

An Example with Loops!

```c
#include<stdio.h>

int main(){
        int x = 0;
        for(int i=0;i<10;i++){
                x = x + 1;
        }
        return x;
}
```

```
main:
        pushq   %rbp
        movq    %rsp, %rbp
        movl    $0, -8(%rbp)
        movl    $0, -4(%rbp)
        jmp     .L2
.L3:
        addl    $1, -8(%rbp)
        addl    $1, -4(%rbp)
.L2:
        cmpl    $9, -4(%rbp)
        jle     .L3
        movl    -8(%rbp), %eax
        popq    %rbp
        ret
```

System Calls in Assembly

```
kernel:
        int     80h    ; //Call kernel
        ret

open:
        push  dword mode
        push  dword flags
        push  dword path
        mov   eax, 5
        call  kernel
        add   esp, byte 12
        ret
```

System Calls in Assembly

```
kernel:
        int     80h    ; //Call kernel
        ret

open:
        push  dword mode
        push  dword flags
        push  dword path
        mov   eax, 5
        call  kernel
        add   esp, byte 12
        ret
```

Syscall Number

A bit different!

A simple fork program →

```
movl    $0, %eax
call    printf
call    fork
```

Embedding Assembly in C

__asm__(“instruction 1”, “instruction 2”, …)

Example:

```
__asm__(
        "movl %edx, %eax\n\t"
        "addl $2, %eax\n\t"
       );
```

Embedding Assembly in C

```c
#include <stdio.h>

int main(int argc, char *argv[]) {

  int x = 5;

  printf("x = %d\n", x);

  __asm__("add $10, %0":"=m"(x));

  printf("x = %d\n", x);

  return 0;
}
```

```
x = 5
x = 15
```

Where will I use assembly?

Where will I use assembly?

- To write Compilers and Device Drivers
- To write viruses and for malware analysis
- Used while programming Real Time Embedded systems
- Implementing Locks for Concurrency.
  We will cover this in the third module of the course!

References

- Chapter 11. x86 Assembly Language Programming, FreeBSD, https://www.freebsd.org/doc/en_US.ISO8859-1/books/developers-handbook/x86.html
- Easy x86-64, http://ian.seyler.me/easy_x86-64/
- Introduction to the GNU/Linux assembler and linker for Intel Pentium processors, https://www.cs.usfca.edu/~cruse/cs210s07/lesson01.ppt
- Is there a way to insert assembly code into C?, https://stackoverflow.com/questions/61341/is-there-a-way-to-insert-assembly-code-into-c

More Topics

- GCC
- Clang
- GCC vs Clang
  - (More: https://opensource.apple.com/source/clang/clang-23/clang/tools/clang/www/comparison.html)
- Debugger
- How to use a debugger (the practical way)

GCC

- The GNU Compiler Collection (GCC) is a compiler system produced by the GNU Project supporting various programming languages.
- GCC is a key component of the GNU toolchain and the standard compiler for most projects related to GNU and Linux, including the Linux kernel

Clang

- Clang is a compiler front end for the C, C++, Objective-C and Objective-C++ programming languages, as well as the OpenMP, OpenCL, RenderScript, CUDA ...

- It uses the LLVM compiler infrastructure as its back end and has been part of the LLVM release cycle since LLVM 2.6

LLVM

- The LLVM compiler infrastructure project is a set of compiler and toolchain technologies, which can be used to develop a front end for any programming language and a back end for any instruction set architecture.

- LLVM is designed around a language-independent intermediate representation (IR) that serves as a portable, high-level assembly language that can be optimized with a variety of transformations over multiple passes.

Clang vs GCC [1/4]

Benefits of Using GCC

- supports languages that clang does not aim to, such as Java, Ada, FORTRAN, etc.
- front-ends are very mature and already support C++. clang's support for C++ is nowhere near what GCC supports.
- supports more targets than LLVM.
- popular and widely adopted.
- GCC does not require a C++ compiler to build it.

Clang vs GCC [1/4]

Benefits of Clang

- The Clang ASTs and design are intended to be easily understandable by anyone who is familiar with the languages involved and who has a basic understanding of how a compiler works.
- Clang is designed as an API from its inception, allowing it to be reused by source analysis tools, refactoring, IDEs (etc) as well as for code generation

Clang vs GCC [1/4]

Benefits of Clang

- Clang aims to provide extremely clear and concise diagnostics (error and warning messages), and includes support for expressive diagnostics.
- Clang inherits a number of features from its use of LLVM as a backend, including support for a bytecode representation for intermediate code, pluggable optimizers, link-time optimization support, Just-In-Time compilation, ability to link in multiple code generators, etc.

Clang vs GCC [1/4]

```
glint@Mark4: ~
  GNU nano 4.8
#include <bits/stdc++.h>

using namespace std

int main(){

        cout<<"Hello World"<<endl;

}
```

```
glint@Mark4:~$ gcc main.cpp
main.cpp:3:20: error: expected ‘;’ before ‘int’
    3 | using namespace std
      |                    ^
      |                    ;
    4 |
    5 | int main(){
      | ~~~
glint@Mark4:~$ clang main.cpp
main.cpp:3:20: error: expected ';' after namespace name
using namespace std
                   ^
                   ;
1 error generated.
glint@Mark4:~$
```

Debugger

- A debugger or debugging tool is a computer program used to test and debug other programs (the "target" program).
- The main use of a debugger is to run the target program under controlled conditions
  - that permit the programmer to track its operations in progress
  - monitor changes in computer resources
  - most often memory areas used by the target program or the computer's operating system that may indicate malfunctioning code.

GNU GDB

- The GNU Debugger (GDB) is a portable debugger that runs on many Unix-like systems and works for many programming languages, including Ada, C, C++, Objective-C, Free Pascal, Fortran, Go ...

How to use GNU GDB (the impractical basics)

- https://u.osu.edu/cstutorials/2018/09/28/how-to-debug-c-program-using-gdb-in-6-simple-steps/
- Might end up as interview questions!

How to use GNU GDB (the practical basics)

- Use an IDE or an advanced text editor with support for debugging
- Something that works on almost all OS and is open source and user extensible is Visual Studio Code.
    - Requirements
        - The source code Eg. main.cpp
        - A compiler Eg gcc
        - A debugger like gdb
        - An editor with support for debugging like Visual Studio Code.

main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code

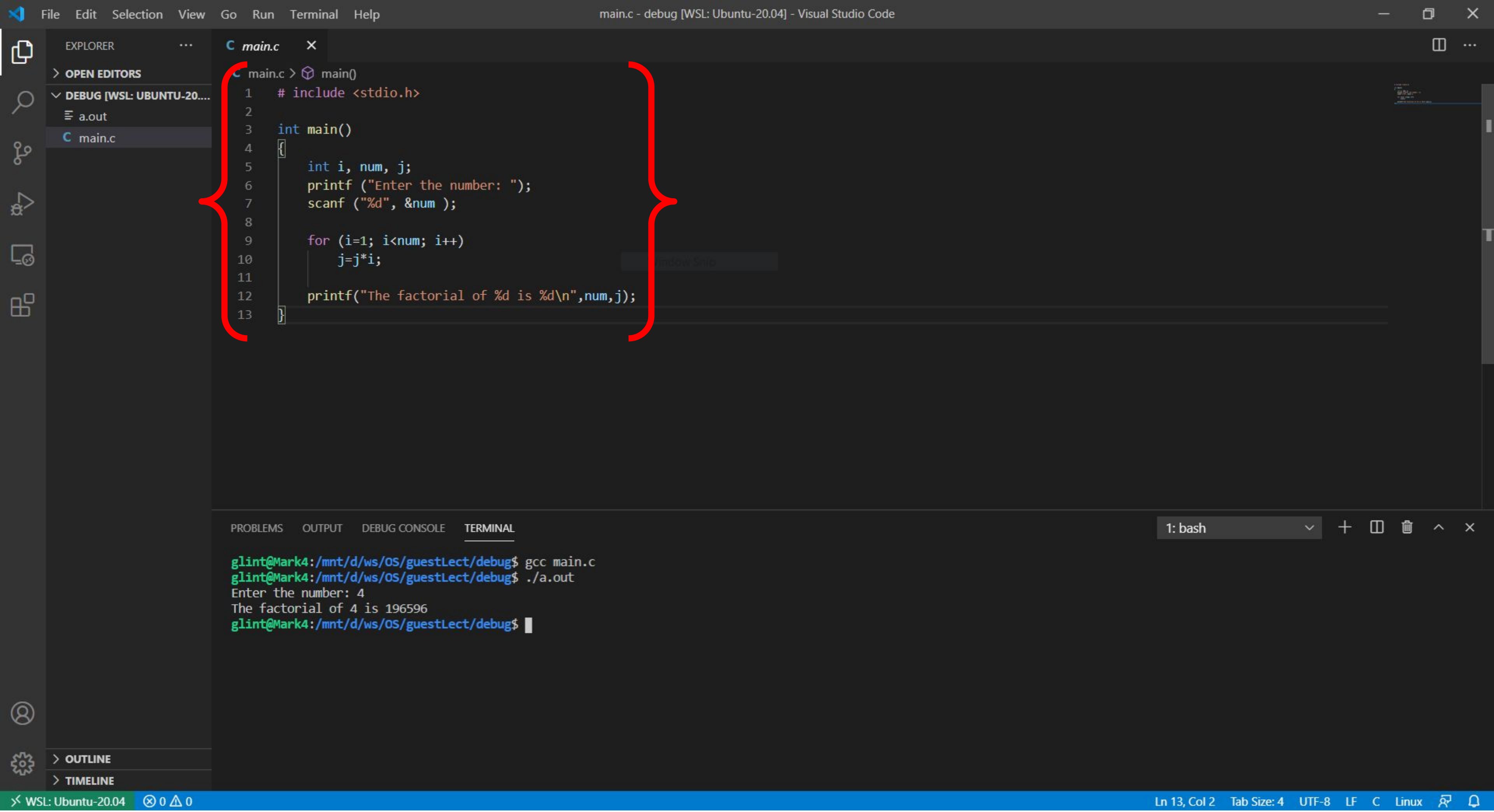
```c
# include <stdio.h>

int main()
{
    int i, num, j;
    printf ("Enter the number: ");
    scanf ("%d", &num );

    for (i=1; i<num; i++)
        j=j*i;

    printf("The factorial of %d is %d\n",num,j);
}
```

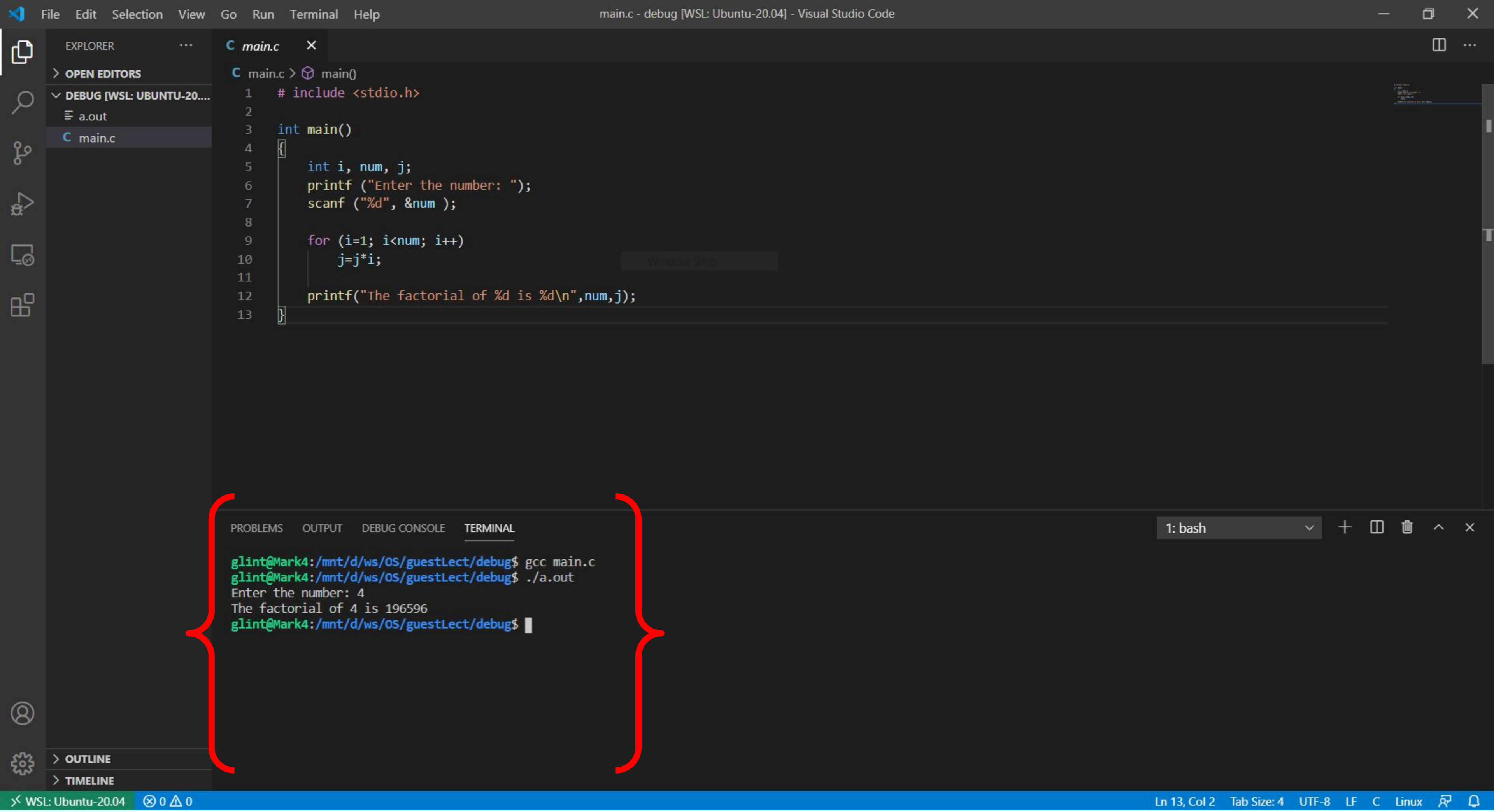
```
glint@Mark4:/mnt/d/ws/OS/guestLect/debug$ gcc main.c
glint@Mark4:/mnt/d/ws/OS/guestLect/debug$ ./a.out
Enter the number: 4
The factorial of 4 is 196596
glint@Mark4:/mnt/d/ws/OS/guestLect/debug$
```


main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
File Edit Selection View Go Run Terminal Help
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20....
a.out
main.c
main.c > main()
include <stdio.h>

int main()
{
    int i, num, j;
    printf ("Enter the number: ");
    scanf ("%d", &num);

    for (i=1; i<num; i++)
        j=j*i;

    printf("The factorial of %d is %d\n",num,j);
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
1: bash
glint@Mark4:/mnt/d/ws/OS/guestLect/debug$ gcc main.c
glint@Mark4:/mnt/d/ws/OS/guestLect/debug$ ./a.out
Enter the number: 4
The factorial of 4 is 196596
glint@Mark4:/mnt/d/ws/OS/guestLect/debug$
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 13, Col 2 Tab Size: 4 UTF-8 LF C Linux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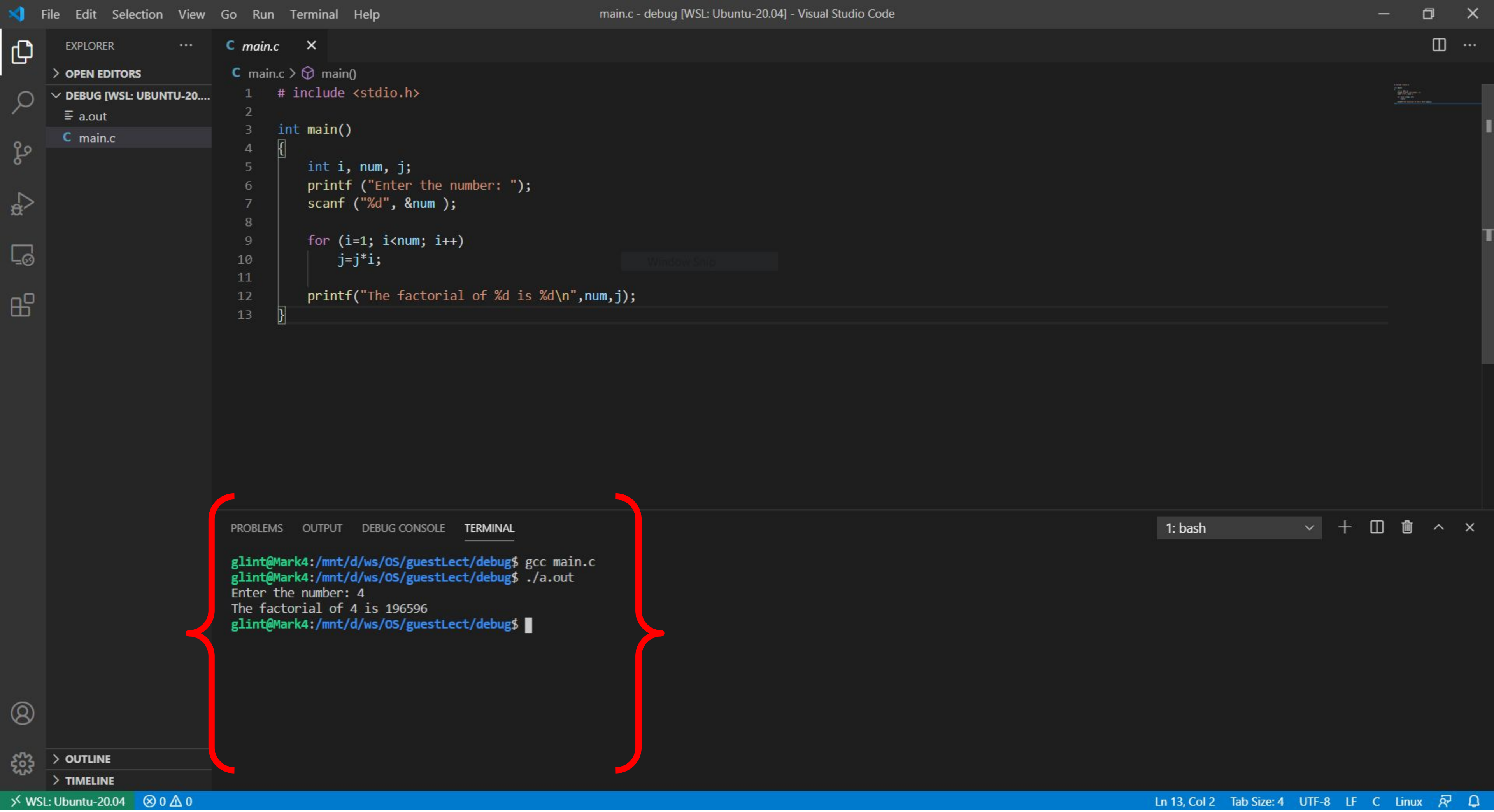
Ln 13, Col 2 Tab Size: 4 UTF-8 LF C Linux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DEBUG [WSL: UBUNTU-20....
a.out
main.c
main.c
main.c > main()
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int main()
{
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printf ("Enter the number: ");
scanf ("%d", &num);
for (i=1; i<num; i++)
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}
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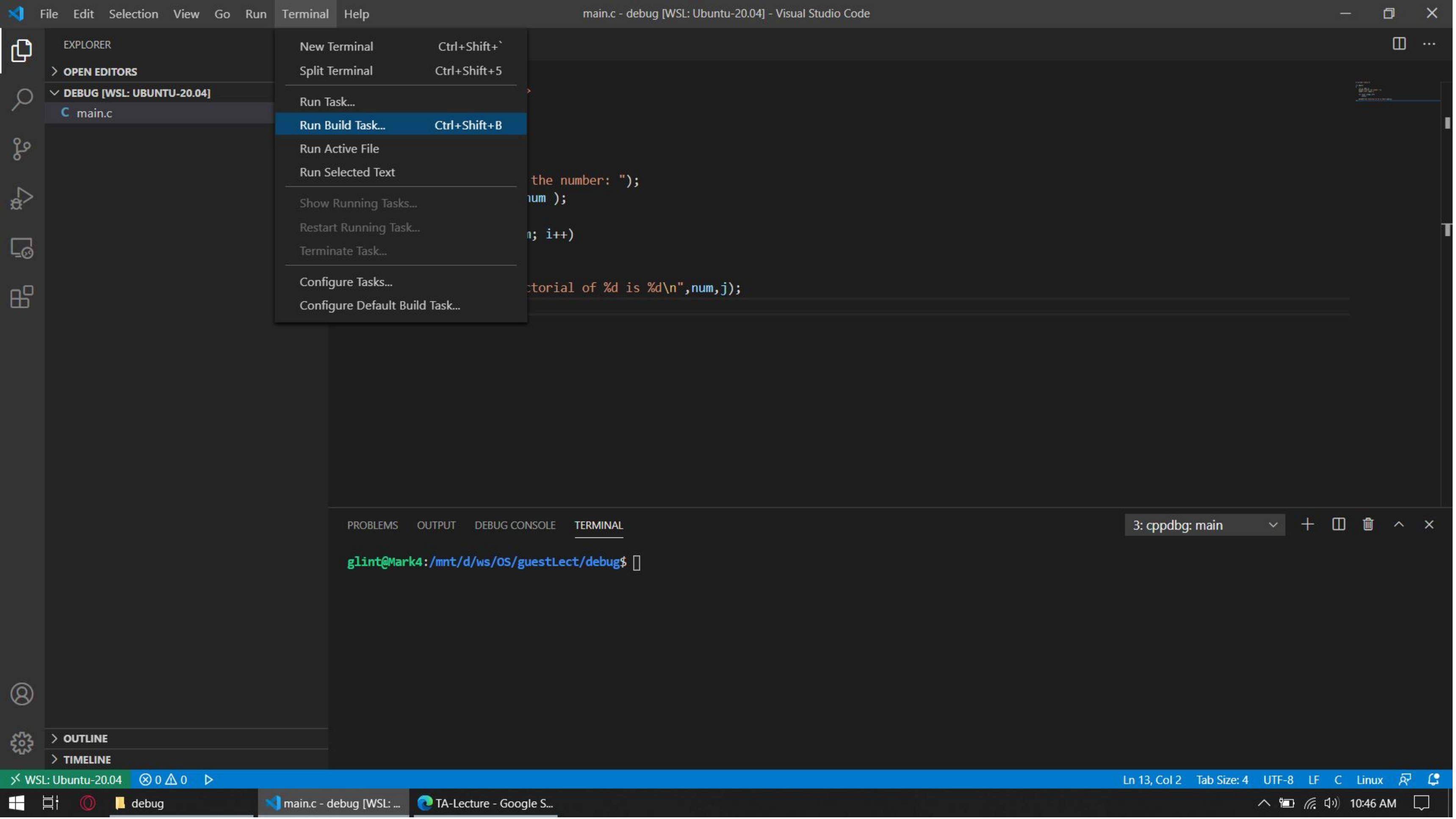
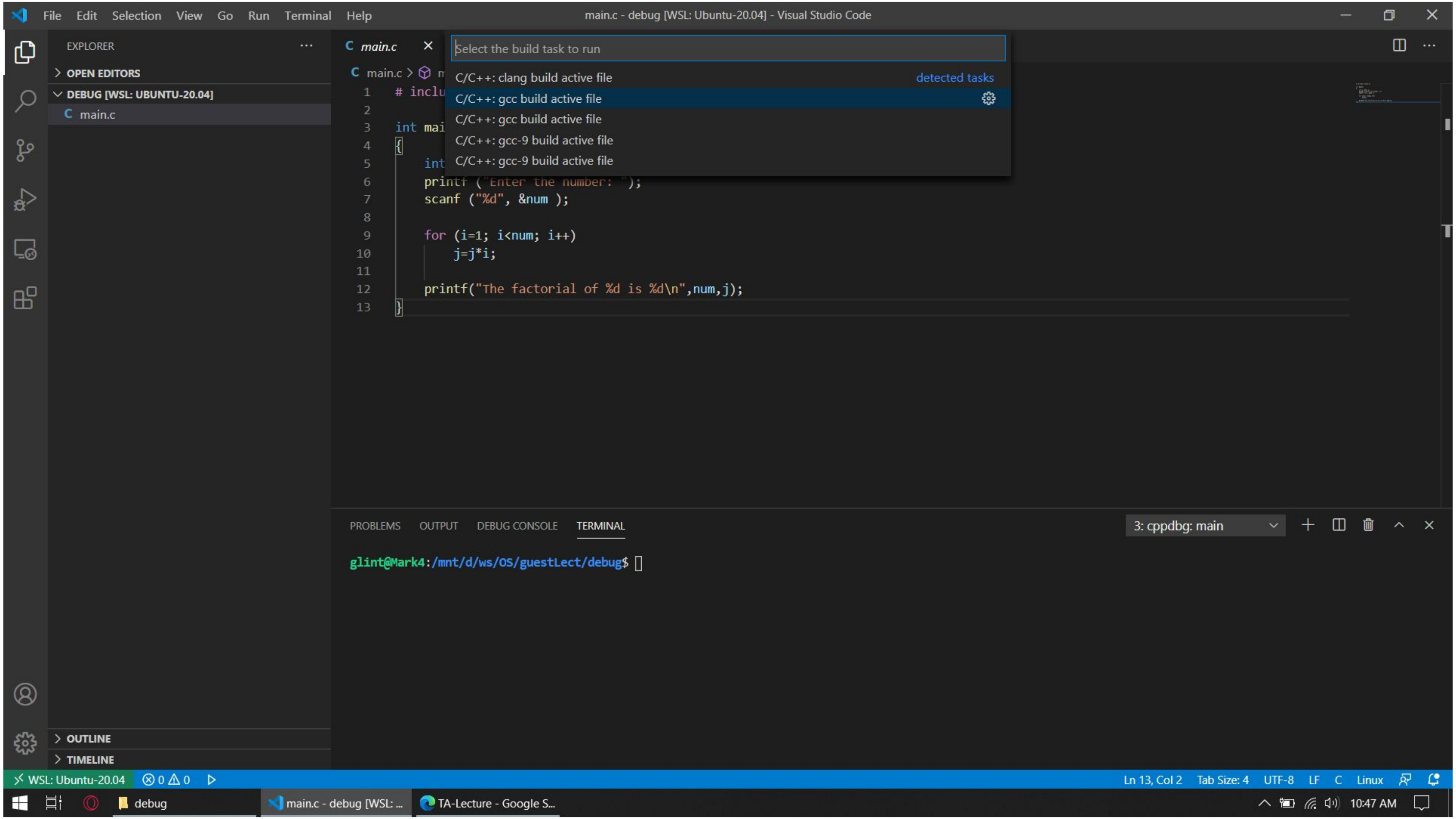
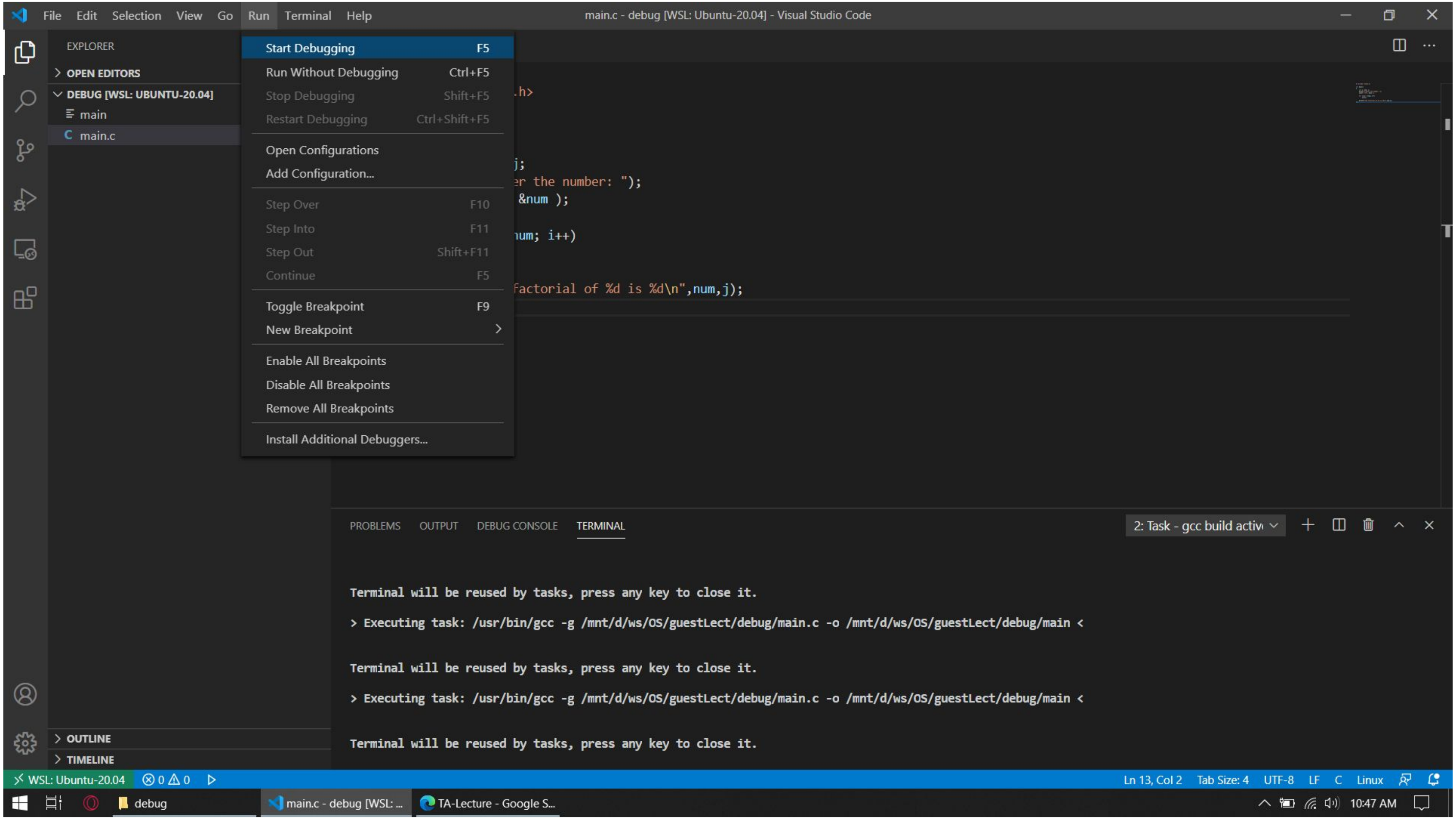
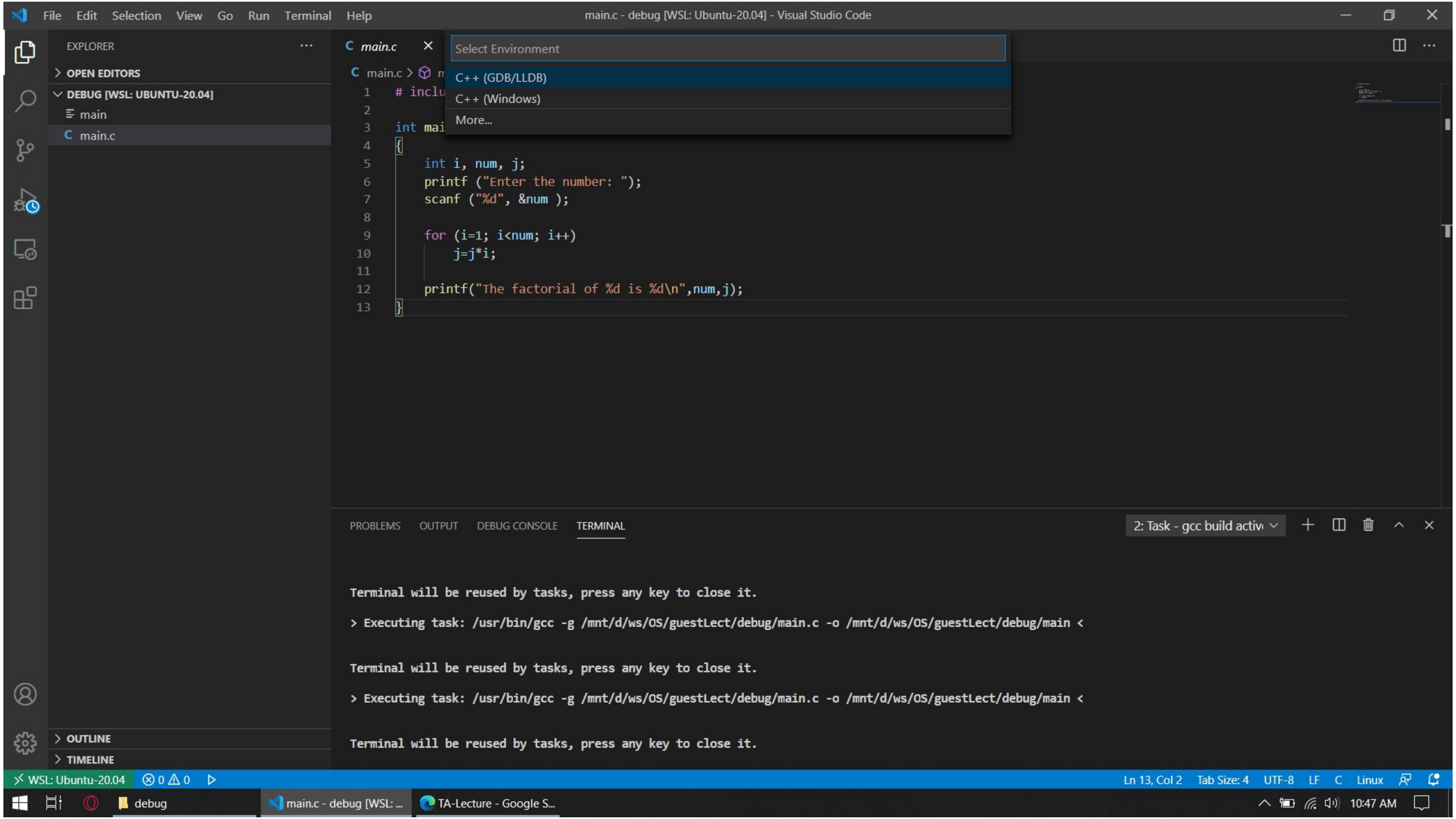
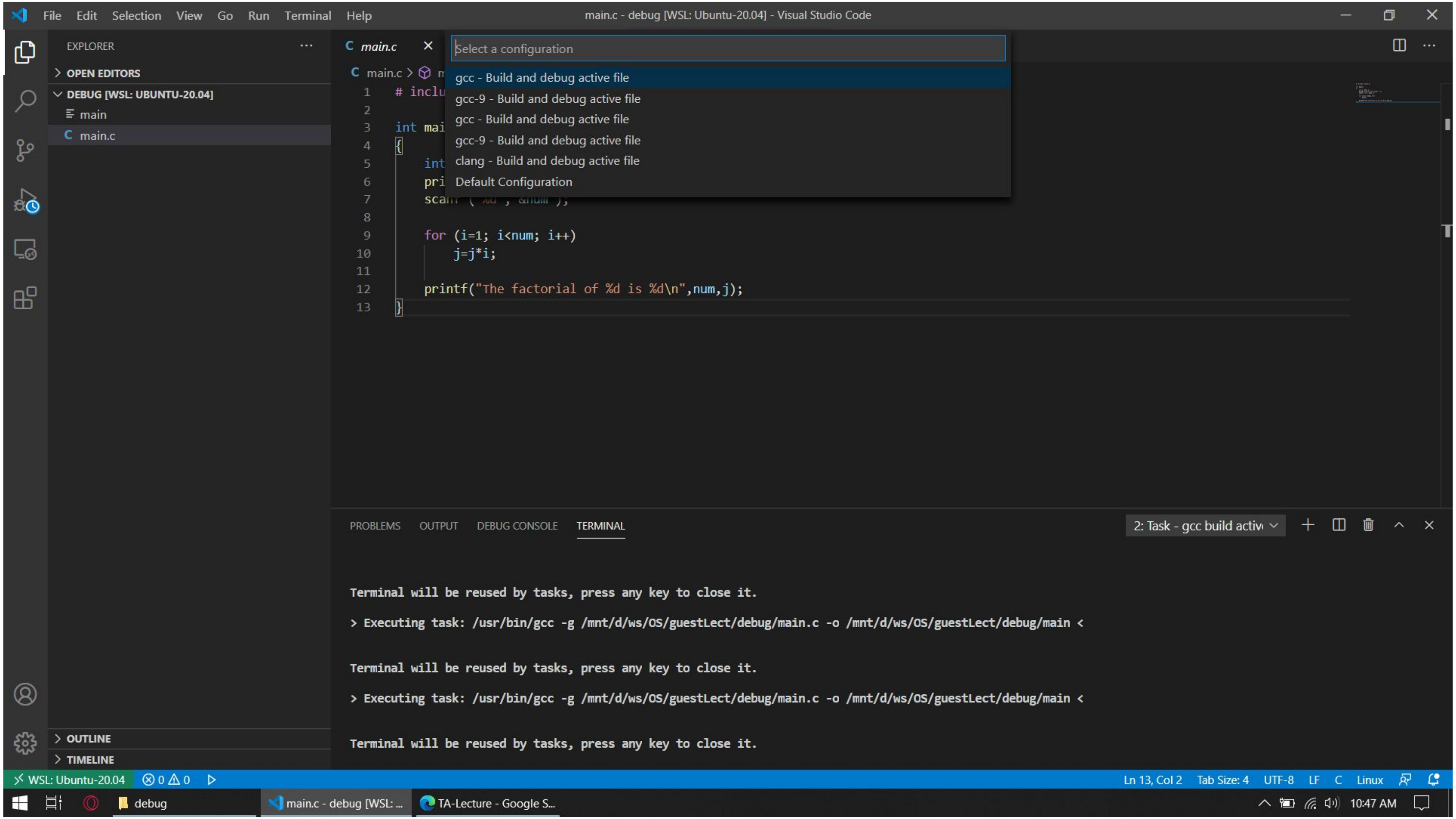
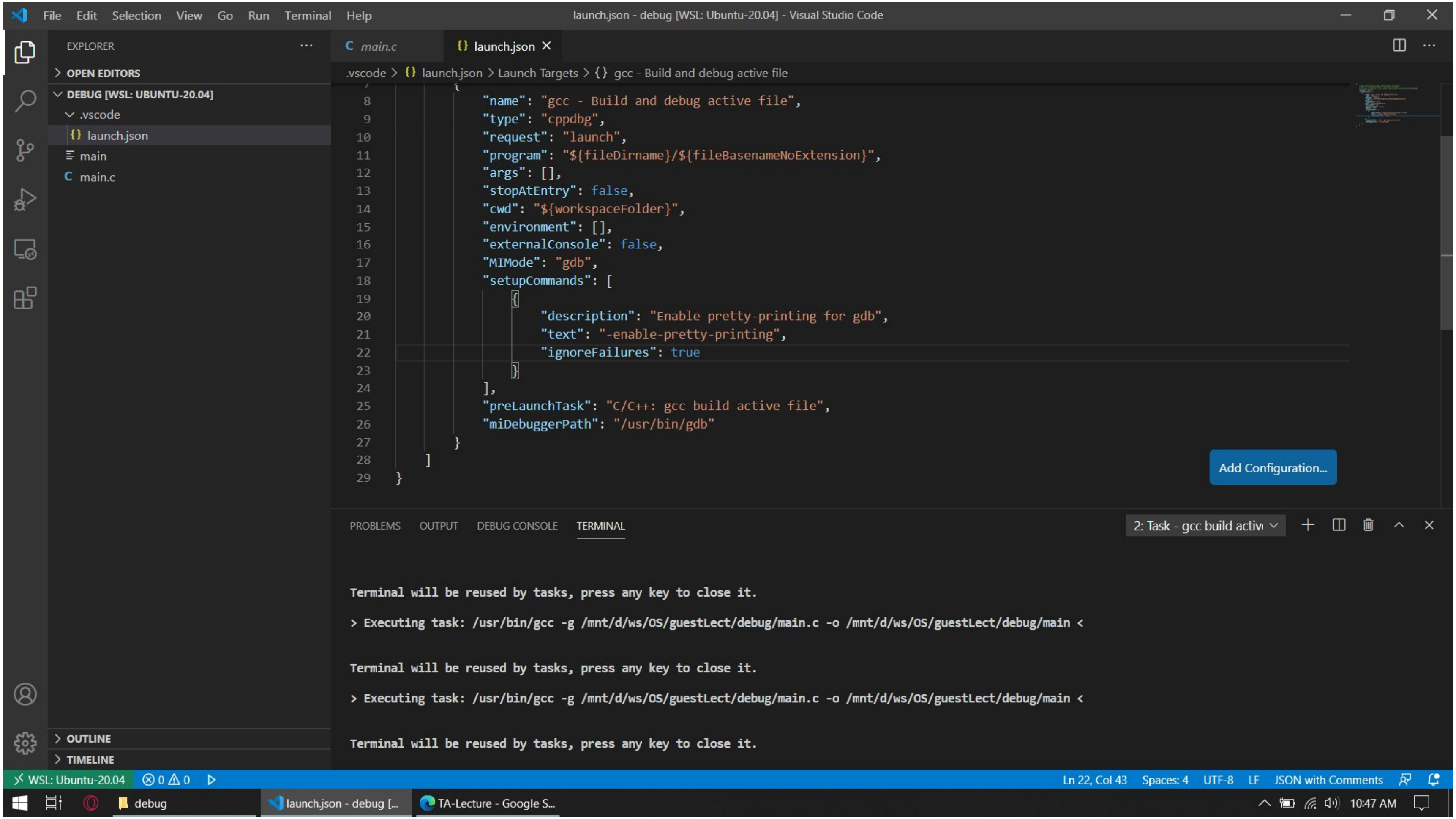

File Edit Selection View Go Run Terminal Help
main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20.04]
main.c
New Terminal Ctrl+Shift+`
Split Terminal Ctrl+Shift+5
Run Task...
Run Build Task... Ctrl+Shift+B
Run Active File
Run Selected Text
Show Running Tasks...
Restart Running Task...
Terminate Task...
Configure Tasks...
Configure Default Build Task...
the number: ");
num);
n; i++)
ctorial of %d is %d\n",num,j);
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
3: cppdbg: main
glint@Mark4:/mnt/d/ws/OS/guestLect/debug$
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 13, Col 2 Tab Size: 4 UTF-8 LF C Linux
debug
main.c - debug [WSL: ...
TA-Lecture - Google S...
10:46 AM

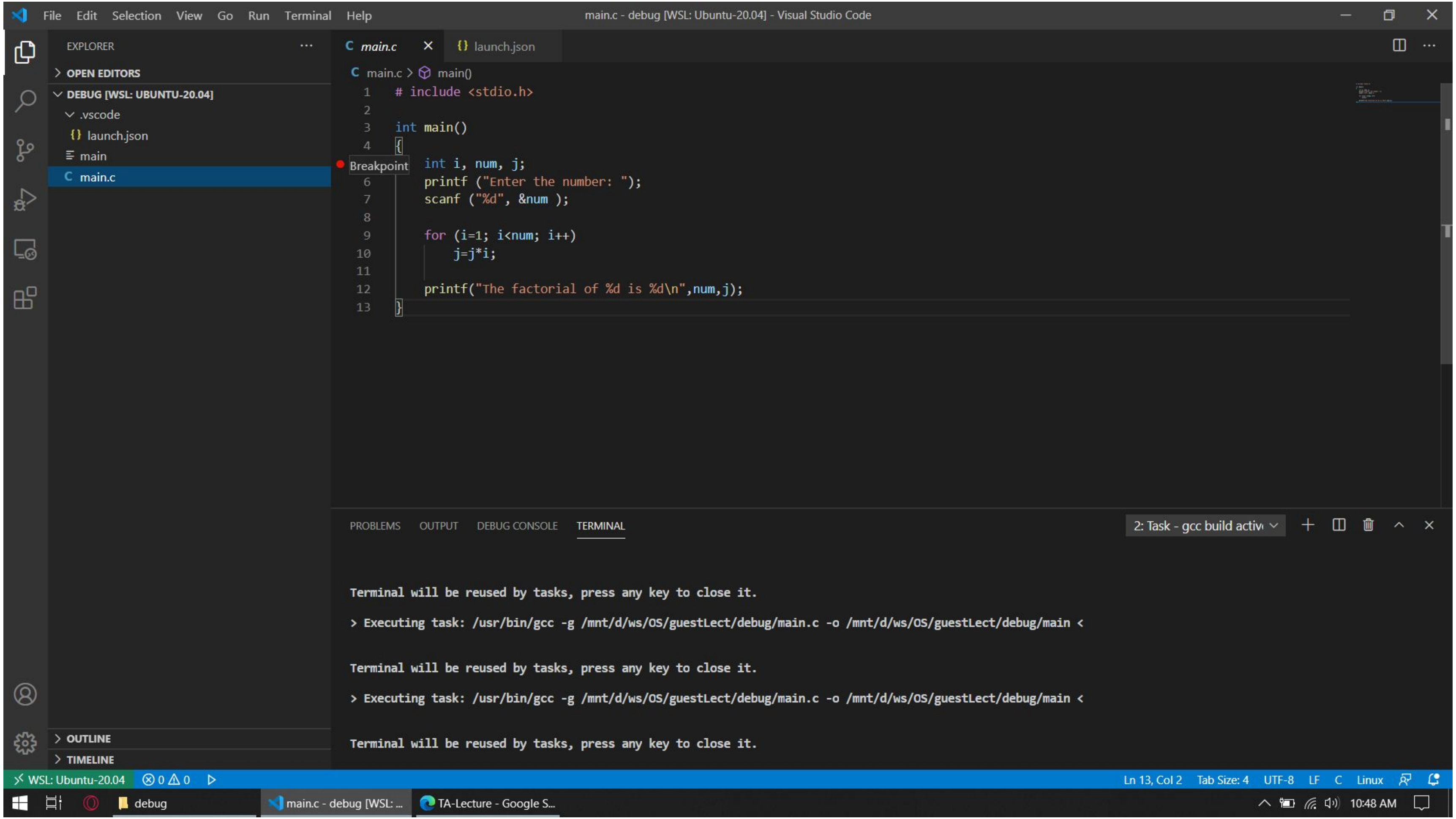

File Edit Selection View Go Run Terminal Help
main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20.04]
main.c
Select the build task to run
C/C++: clang build active file
detected tasks
C/C++: gcc build active file
C/C++: gcc build active file
C/C++: gcc-9 build active file
C/C++: gcc-9 build active file
printf ("Enter the number: ");
scanf ("%d", &num);
for (i=1; i<num; i++)
j=j*i;
printf("The factorial of %d is %d\n",num,j);
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
3: cppdbg: main
glint@Mark4:/mnt/d/ws/OS/guestLect/debug$
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 13, Col 2 Tab Size: 4 UTF-8 LF C Linux
debug
main.c - debug [WSL: ...
TA-Lecture - Google S...
10:47 AM

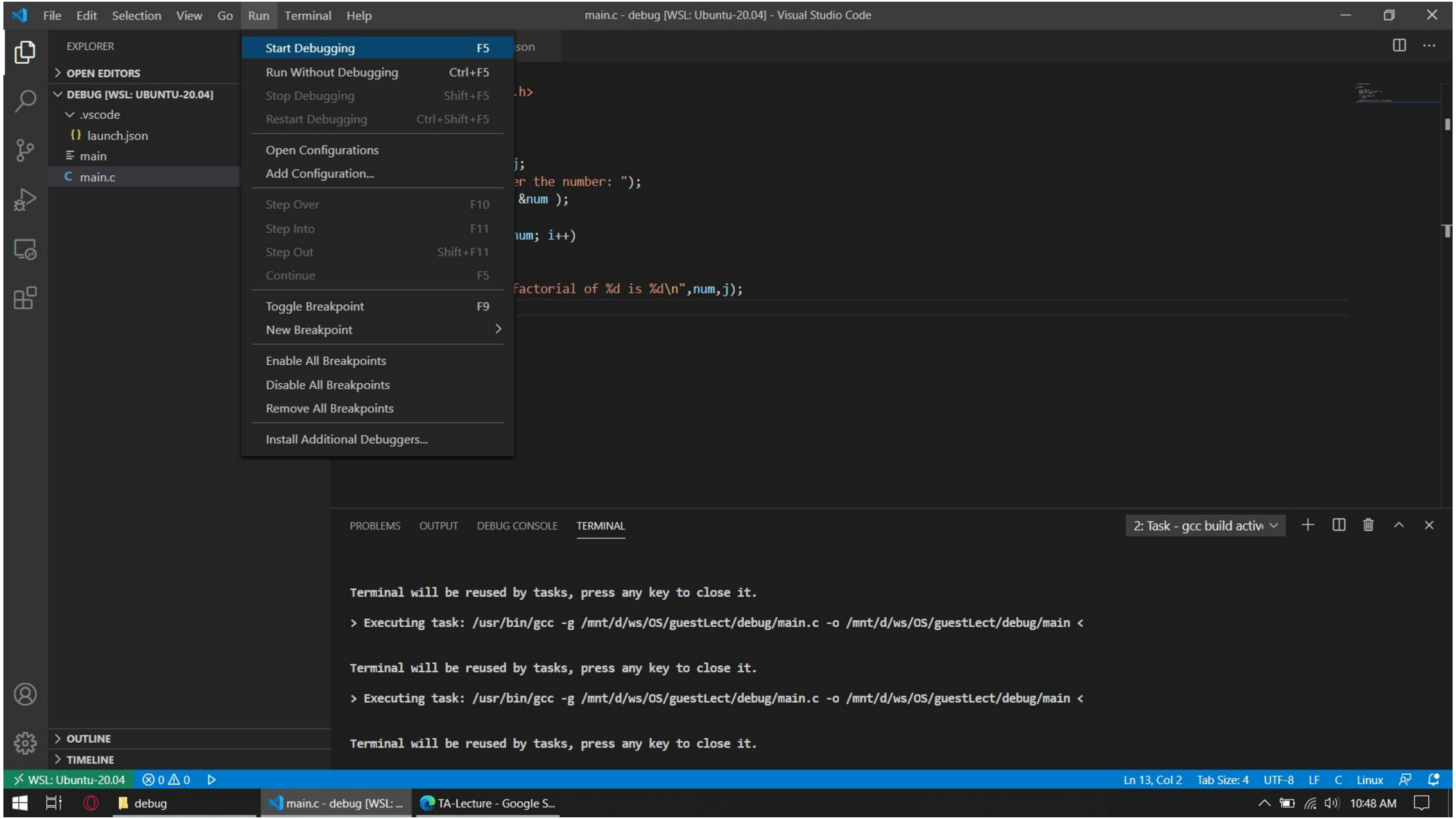

File Edit Selection View Go Run Terminal Help
main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20.04]
main
main.c
Start Debugging F5
Run Without Debugging Ctrl+F5
Stop Debugging Shift+F5
Restart Debugging Ctrl+Shift+F5
Open Configurations
Add Configuration...
Step Over F10
Step Into F11
Step Out Shift+F11
Continue F5
Toggle Breakpoint F9
New Breakpoint
Enable All Breakpoints
Disable All Breakpoints
Remove All Breakpoints
Install Additional Debuggers...
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
2: Task - gcc build activ
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 13, Col 2 Tab Size: 4 UTF-8 LF C Linux
debug
main.c - debug [WSL: ...
TA-Lecture - Google S...
10:47 AM

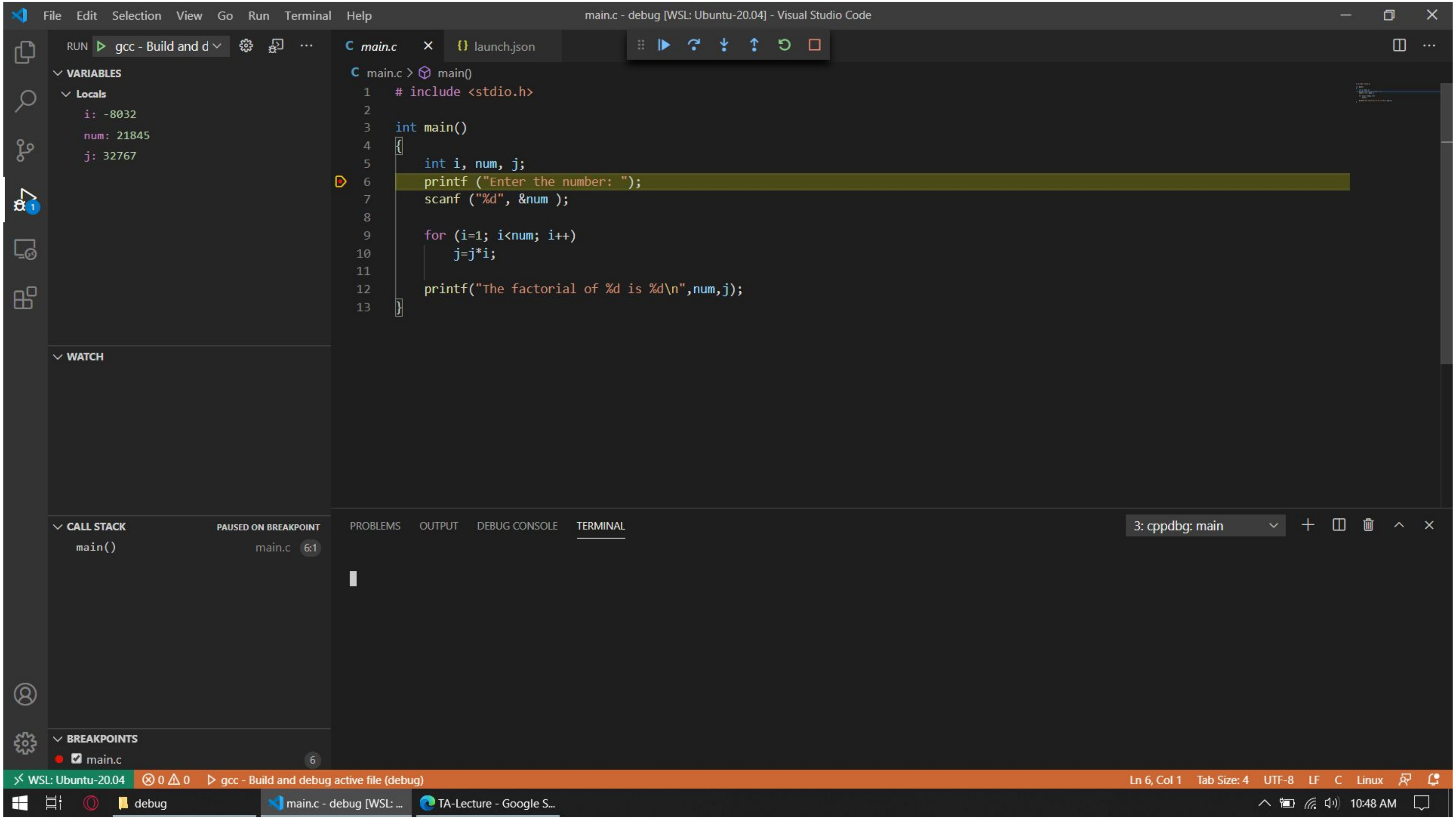

File Edit Selection View Go Run Terminal Help
main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20.04]
main
main.c
main.c
Select Environment
C++ (GDB/LLDB)
C++ (Windows)
More...
inclu
int mai
{
int i, num, j;
printf ("Enter the number: ");
scanf ("%d", &num);
for (i=1; i<num; i++)
j=j*i;
printf("The factorial of %d is %d\n",num,j);
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
2: Task - gcc build activ
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 13, Col 2 Tab Size: 4 UTF-8 LF C Linux
debug
main.c - debug [WSL: ...
TA-Lecture - Google S...
10:47 AM


main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
File Edit Selection View Go Run Terminal Help
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20.04]
main
main.c
Select a configuration
gcc - Build and debug active file
gcc-9 - Build and debug active file
gcc - Build and debug active file
gcc-9 - Build and debug active file
clang - Build and debug active file
Default Configuration
for (i=1; i<num; i++)
j=j*i;
printf("The factorial of %d is %d\n",num,j);
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
2: Task - gcc build activ
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 13, Col 2 Tab Size: 4 UTF-8 LF C Linux
debug
main.c - debug [WSL: ...
TA-Lecture - Google S...
10:47 AM


launch.json - debug [WSL: Ubuntu-20.04] - Visual Studio Code
File Edit Selection View Go Run Terminal Help
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20.04]
.vscode
launch.json
main
main.c
.vscode > launch.json > Launch Targets > gcc - Build and debug active file
"name": "gcc - Build and debug active file",
"type": "cppdbg",
"request": "launch",
"program": "${fileDirname}/${fileBasenameNoExtension}",
"args": [],
"stopAtEntry": false,
"cwd": "${workspaceFolder}",
"environment": [],
"externalConsole": false,
"MIMode": "gdb",
"setupCommands": [
{
"description": "Enable pretty-printing for gdb",
"text": "-enable-pretty-printing",
"ignoreFailures": true
}
],
"preLaunchTask": "C/C++: gcc build active file",
"miDebuggerPath": "/usr/bin/gdb"
}
]
}
Add Configuration...
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
2: Task - gcc build activ
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 22, Col 43 Spaces: 4 UTF-8 LF JSON with Comments
debug
launch.json - debug [...
TA-Lecture - Google S...
10:47 AM


File Edit Selection View Go Run Terminal Help
main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20.04]
.vscode
launch.json
main
main.c
main.c
launch.json
main.c > main()
include <stdio.h>
int main()
{
Breakpoint
int i, num, j;
printf ("Enter the number: ");
scanf ("%d", &num);
for (i=1; i<num; i++)
j=j*i;
printf("The factorial of %d is %d\n",num,j);
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
2: Task - gcc build activ
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 13, Col 2
Tab Size: 4
UTF-8
LF
C
Linux
debug
main.c - debug [WSL: ...
TA-Lecture - Google S...
10:48 AM


File Edit Selection View Go Run Terminal Help
main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
EXPLORER
OPEN EDITORS
DEBUG [WSL: UBUNTU-20.04]
.vscode
launch.json
main
main.c
Start Debugging F5
Run Without Debugging Ctrl+F5
Stop Debugging Shift+F5
Restart Debugging Ctrl+Shift+F5
Open Configurations
Add Configuration...
Step Over F10
Step Into F11
Step Out Shift+F11
Continue F5
Toggle Breakpoint F9
New Breakpoint
Enable All Breakpoints
Disable All Breakpoints
Remove All Breakpoints
Install Additional Debuggers...
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
2: Task - gcc build activ
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
> Executing task: /usr/bin/gcc -g /mnt/d/ws/OS/guestLect/debug/main.c -o /mnt/d/ws/OS/guestLect/debug/main <
Terminal will be reused by tasks, press any key to close it.
OUTLINE
TIMELINE
WSL: Ubuntu-20.04
Ln 13, Col 2 Tab Size: 4 UTF-8 LF C Linux
debug
main.c - debug [WSL: ...
TA-Lecture - Google S...
10:48 AM



main.c - debug [WSL: Ubuntu-20.04] - Visual Studio Code
File Edit Selection View Go Run Terminal Help
RUN gcc - Build and d
main.c
launch.json
VARIABLES
Locals
i: -8032
num: 21845
j: 32767
WATCH
CALL STACK
PAUSED ON BREAKPOINT
main() main.c 6:1
BREAKPOINTS
main.c 6
main.c > main()
include <stdio.h>
int main()
{
int i, num, j;
printf ("Enter the number: ");
scanf ("%d", &num);
for (i=1; i<num; i++)
j=j*i;
printf("The factorial of %d is %d\n",num,j);
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
3: cppdbg: main
WSL: Ubuntu-20.04
gcc - Build and debug active file (debug)
Ln 6, Col 1 Tab Size: 4 UTF-8 LF C Linux
debug
main.c - debug [WSL: ...
TA-Lecture - Google S...
10:48 AM